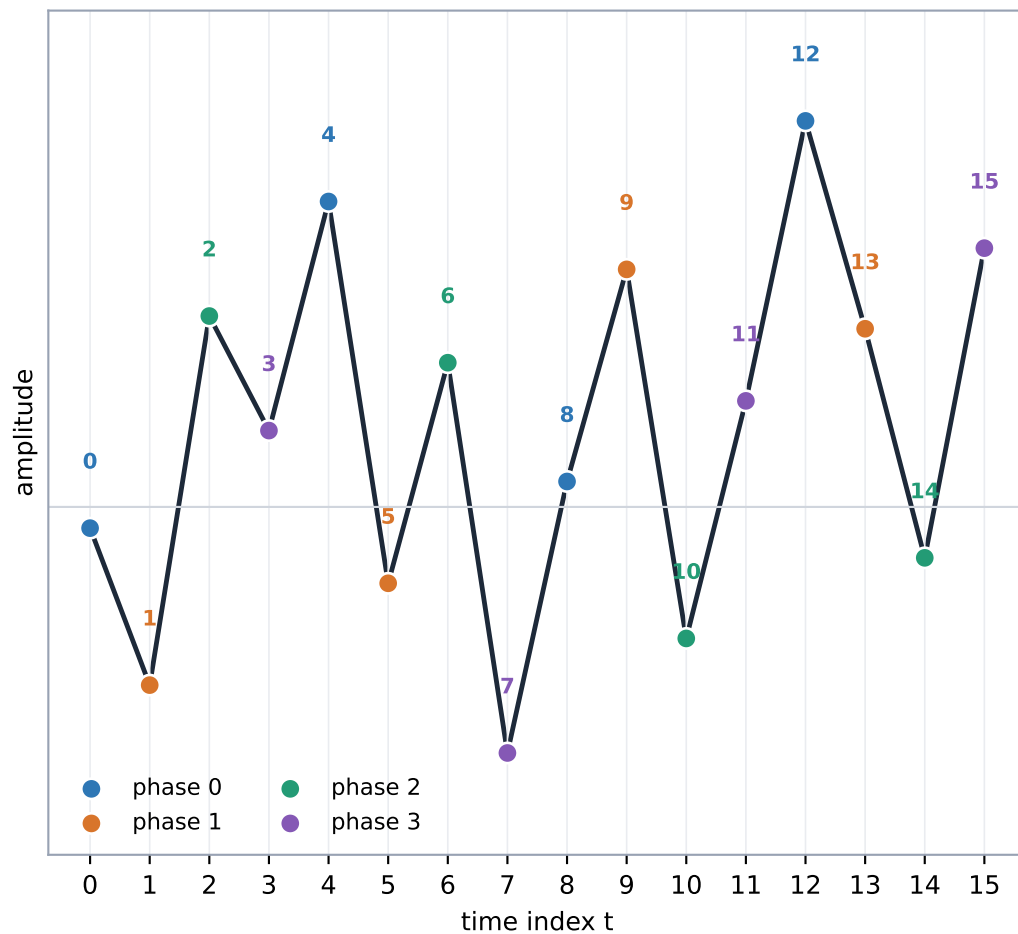
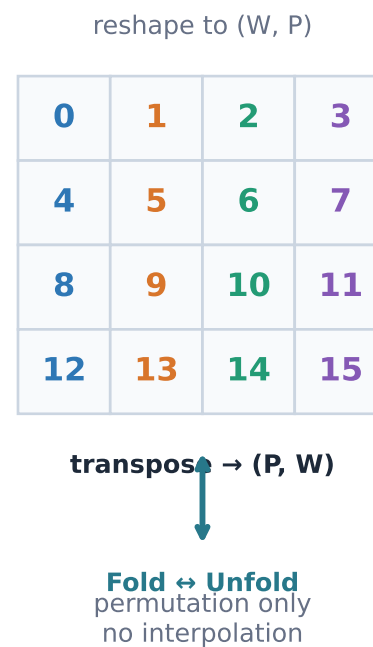


# Phase-Interleaved Lossless Temporal Folding

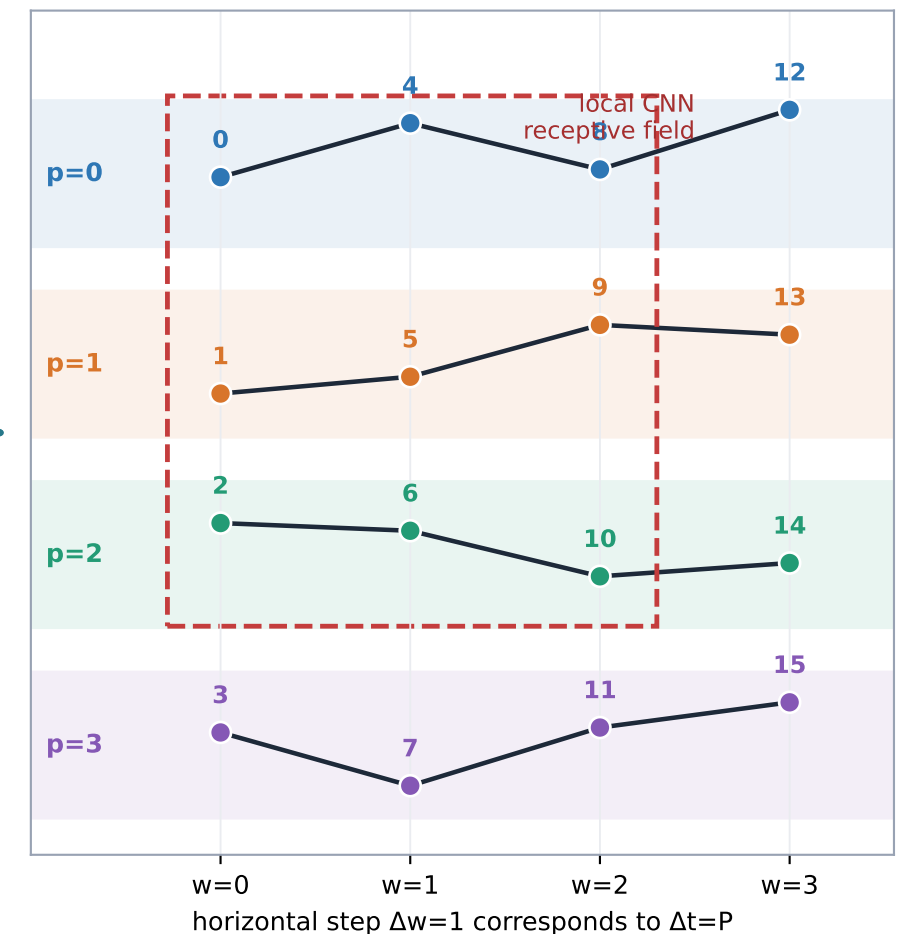
## A. Raw EEG channel



## B. Reindex



## C. Folded EEG rows



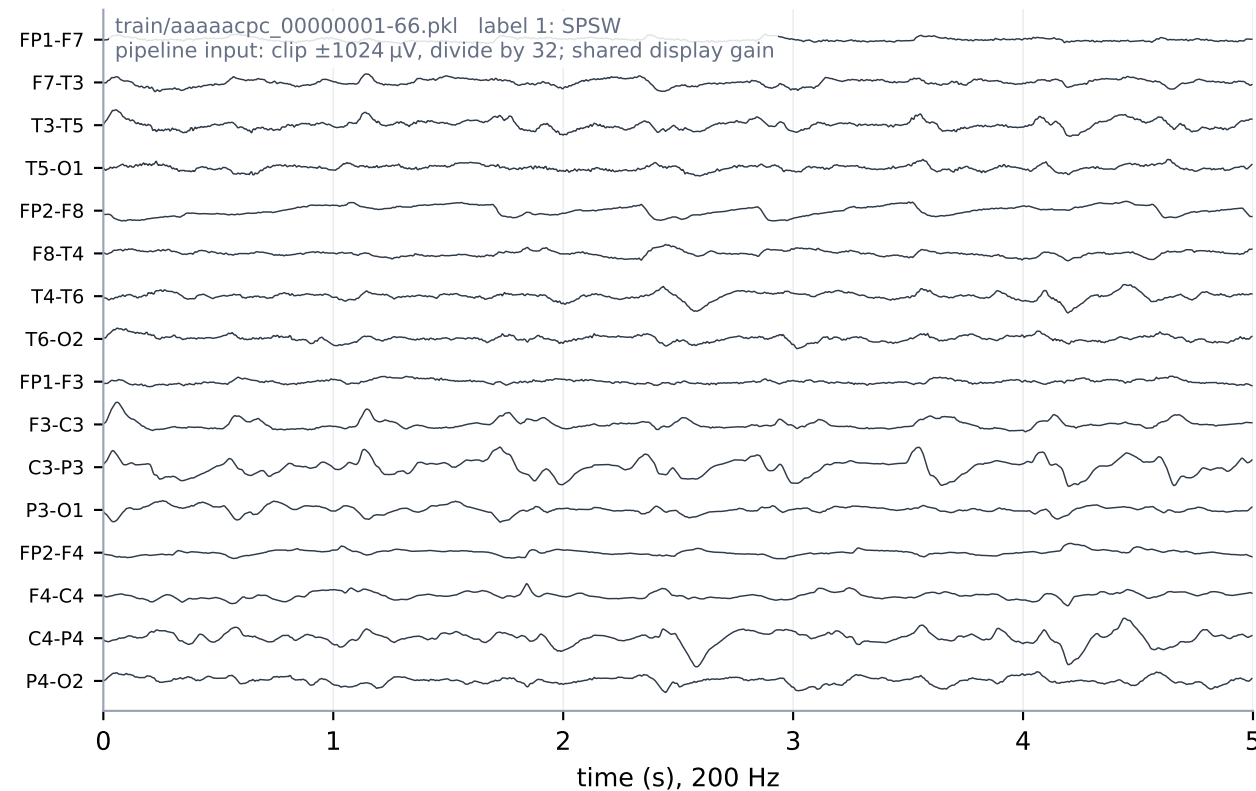
colors repeat every  $P=4$  samples:  $p = t \bmod P$

**Same samples · New adjacency · Zero parameters**

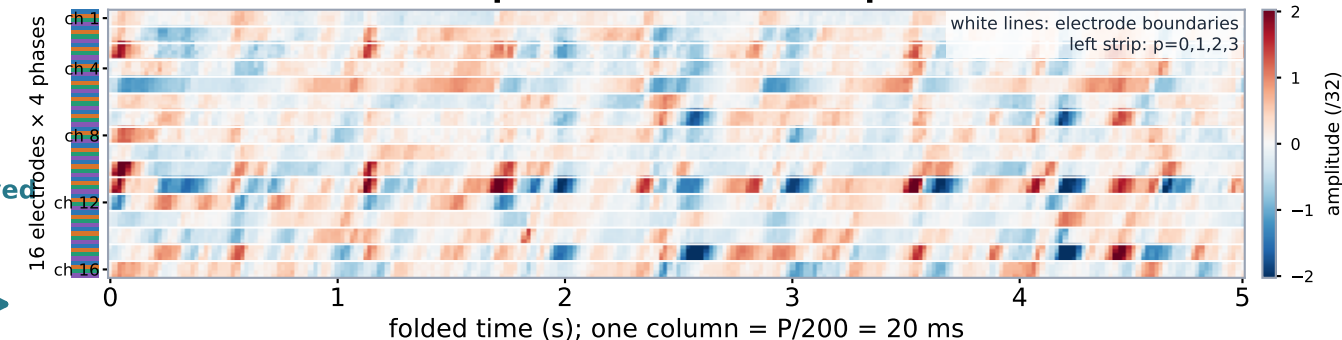
$$I[cP + p, w] = X[c, wP + p]$$

## D. Concrete TUEV example: exact pipeline tensor before and after lossless $P=4$ folding

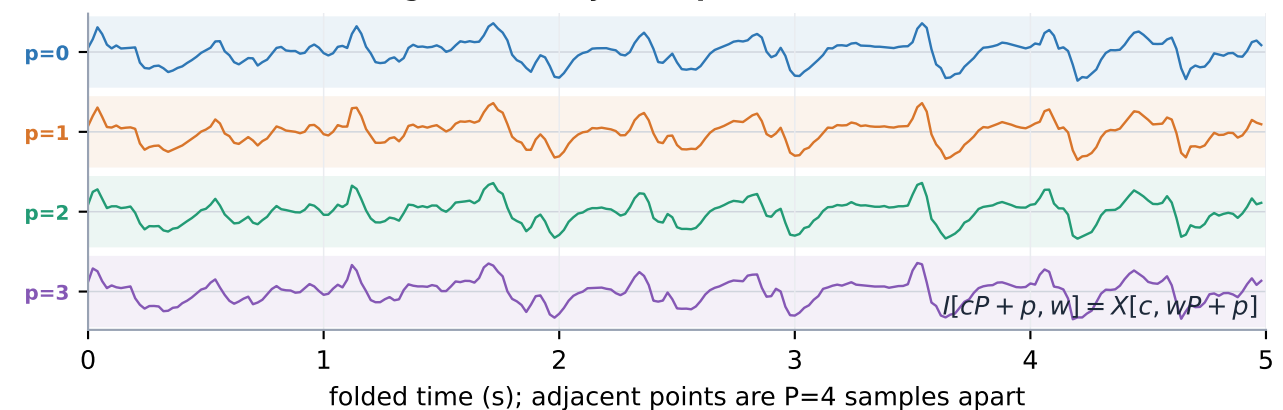
### D1. Real TUEV EEG segment



### D2. Folded tensor: 64 phase rows $\times$ 250 steps



### D3. Real C3-P3 enlarged: four adjacent phase rows



phase-interleaved  
fold,  $P=4$

$16 \times 1000$   
 $\rightarrow$   
 $64 \times 250$

same 16,000 values